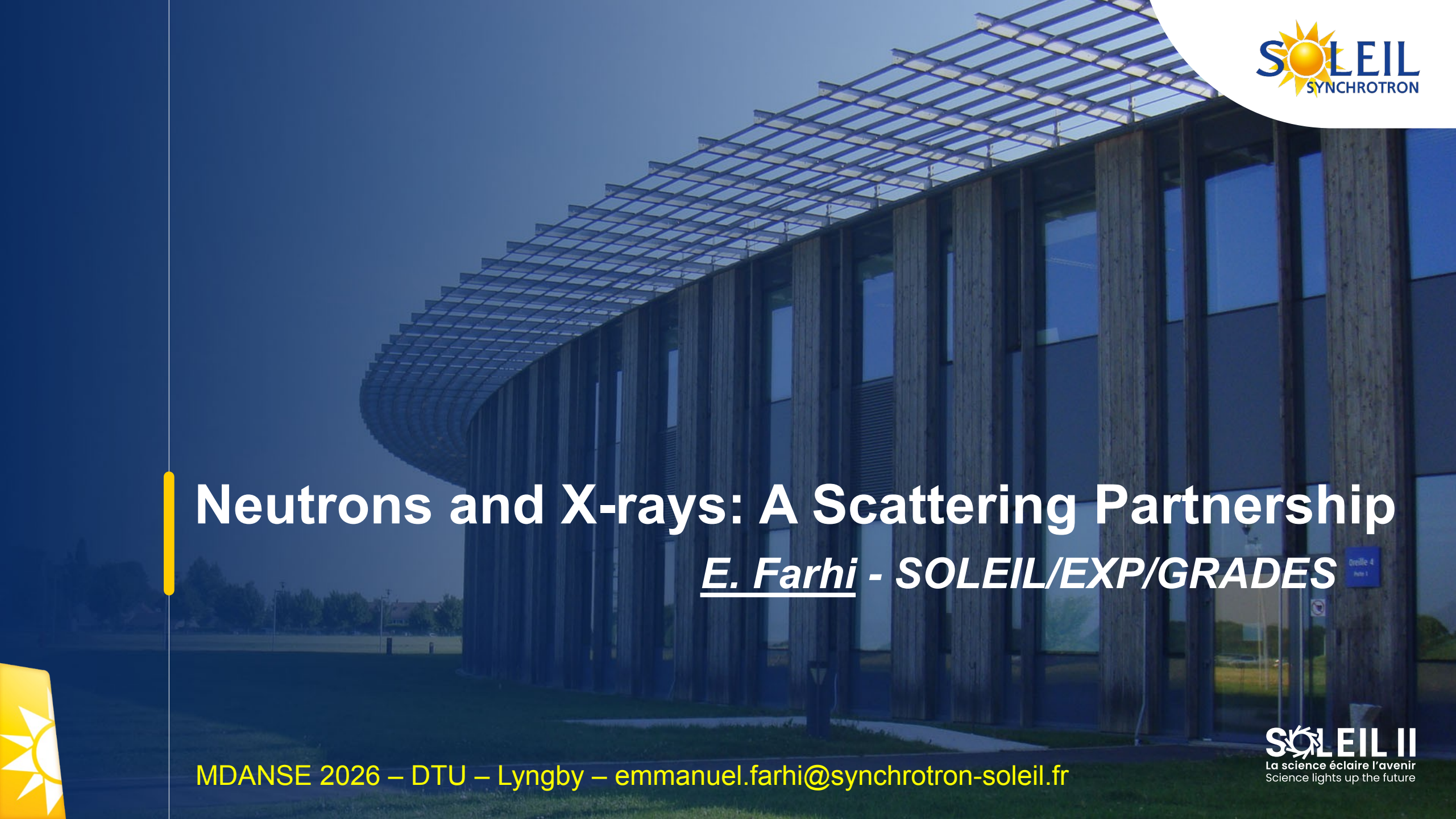


Neutrons and X-rays: A Scattering Partnership

E. Farhi - SOLEIL/EXP/GRADES

MDANSE 2026 – DTU – Lyngby – emmanuel.farhi@synchrotron-soleil.fr

SOLEIL II
La science éclaire l'avenir
Science lights up the future



Some theory

Warning: do not sleep now

SOLEIL II
La science éclaire l'avenir
Science lights up the future

n and X interact with atoms

X-rays “see” the electron cloud surrounding the atomic nucleus – the higher atomic no. ‘**Z**’, the more scattering.

In contrast **neutrons** “see” the nucleus – no simple connection between level of scattering and atomic no.

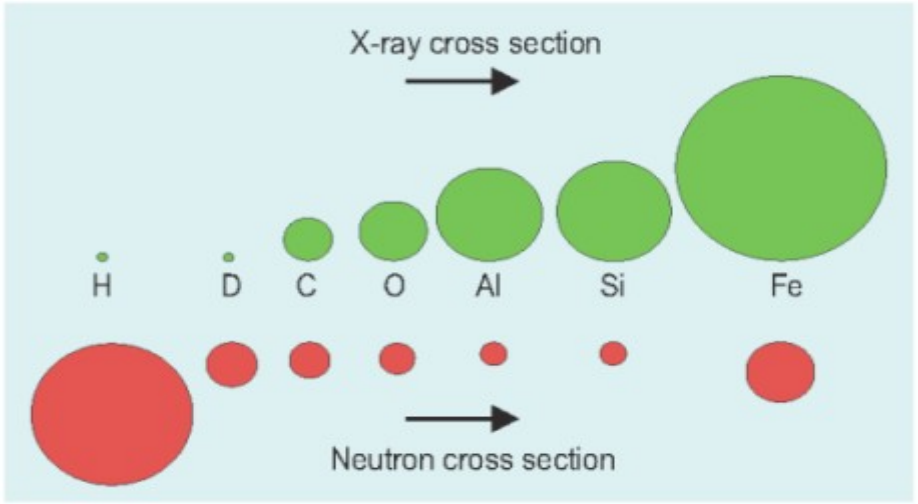
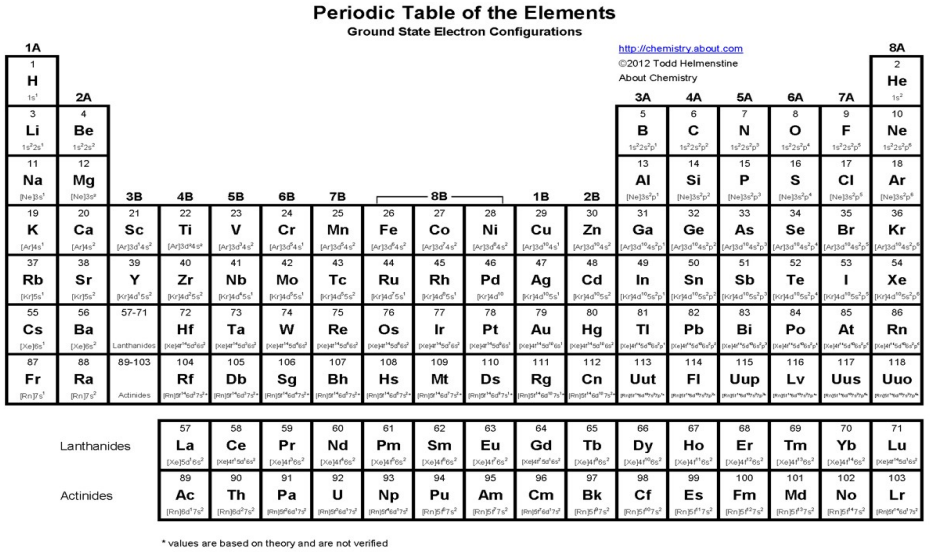


Fig. 2. Neutron and x-ray scattering cross-sections compared. Note that neutrons penetrate through Al much better than x rays do, yet are strongly scattered by hydrogen.



Result: Two very different response/contrasts.

n and X interact with atoms

X-rays “see” the electron cloud surrounding the atomic nucleus – the higher atomic no. ‘**Z**’, the more scattering.

In contrast **neutrons** “see” the nucleus – no simple connection between level of scattering and atomic no.

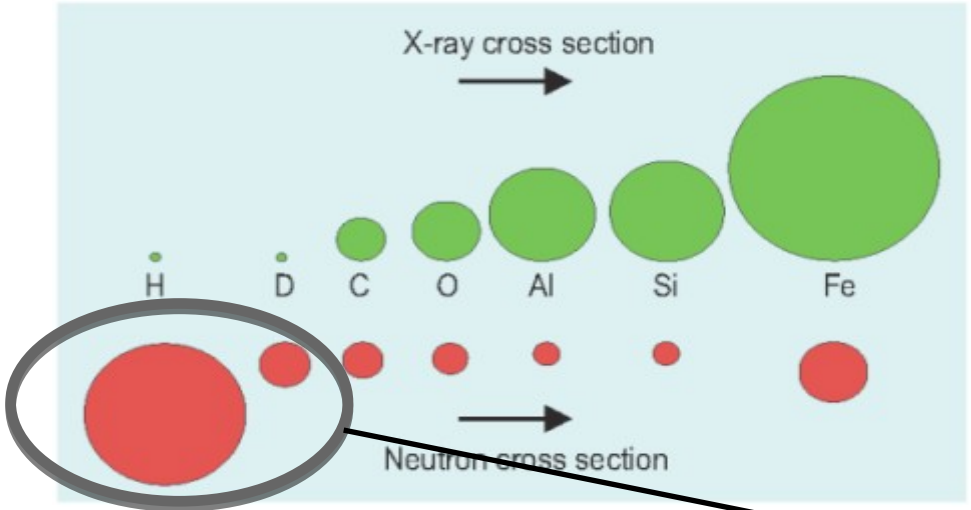
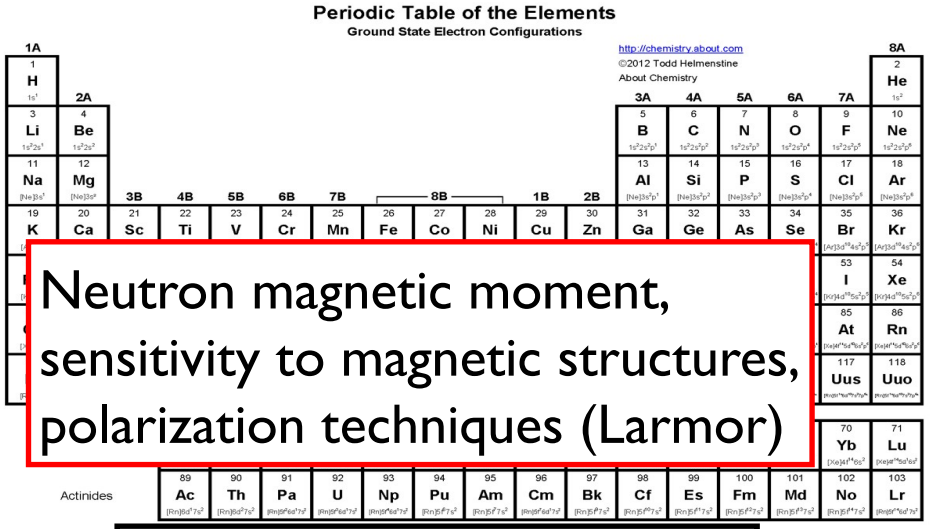


Fig. 2. Neutron and x-ray scattering cross-sections compared. Note that neutrons penetrate through Al much better than x rays do, yet are strongly scattered by hydrogen.



Neutron magnetic moment, sensitivity to magnetic structures, polarization techniques (Larmor)

Large contrast H vs D, ideal for biological systems, via deuteration

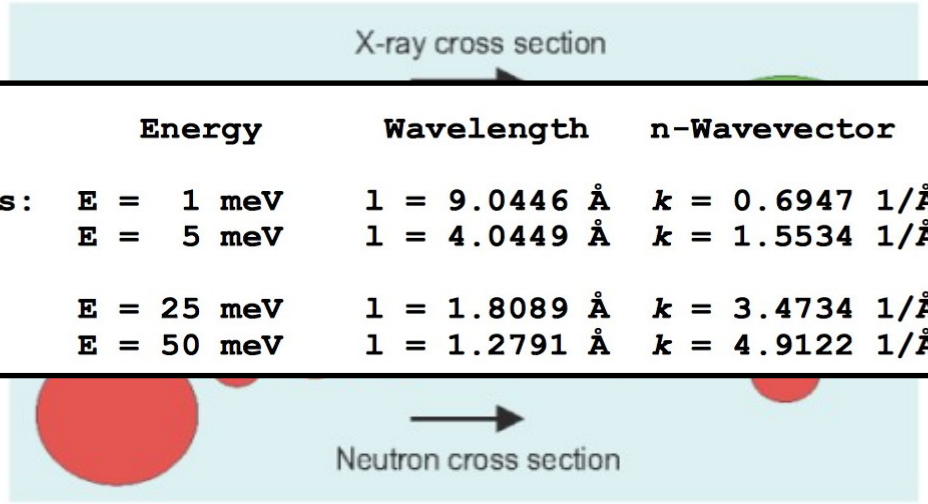
Result: Two very different response/contrasts.

SOLEIL II
La science éclaire l'avenir
Science lights up the future

n and X interact with atoms

X-rays “see” the electron cloud surrounding the atomic nucleus – the higher atomic no. ‘**Z**’, the more scattering.

In contrast **neutrons** “see” the nucleus – no simple connection between level of scattering and atomic no.



	Energy	Wavelength	n-Wavevector	Velocity	Frequency
cold neutrons:	E = 1 meV	l = 9.0446 Å	*k* = 0.6947 1/Å	v = 437 m/s	*v* = 0.2418 THz
	E = 5 meV	l = 4.0449 Å	*k* = 1.5534 1/Å	v = 978 m/s	*v* = 1.2090 THz
thermal neutrons:	E = 25 meV	l = 1.8089 Å	*k* = 3.4734 1/Å	v = 2187 m/s	*v* = 6.045 THz
	E = 50 meV	l = 1.2791 Å	*k* = 4.9122 1/Å	v = 3093 m/s	*v* = 12.090 THz

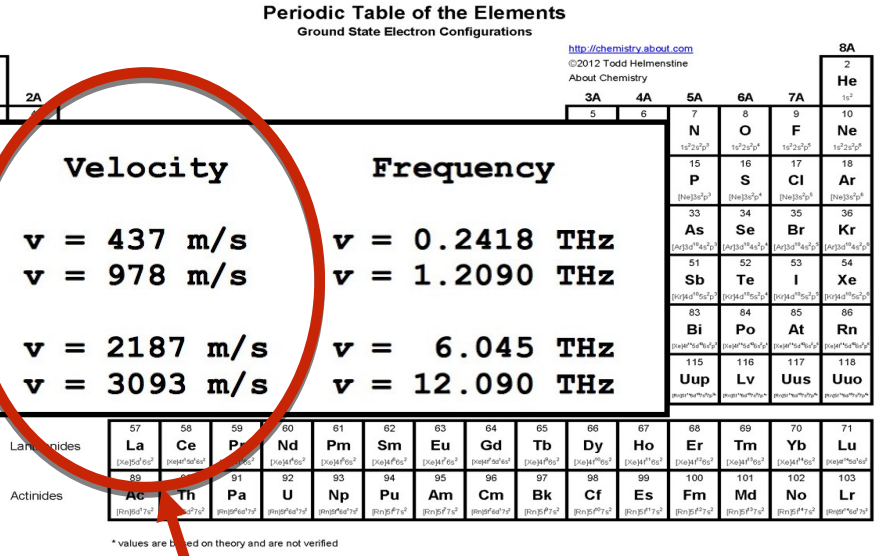


Fig. 2. Neutron and x-ray scattering cross-sections compared. Note that neutrons penetrate through Al much better than x rays do, yet are strongly scattered by hydrogen.

Low velocities makes beam-tailoring using mechanical devices feasible

Result: Two very different response/contrasts.

SOLEIL II
La science éclaire l’avenir
Science lights up the future

X ray and neutron cross sections

The atomic scattering factor/cross section:

$b \leftrightarrow r_e.f$

n $\sigma \leftrightarrow r_e^2.f^2$ **X**

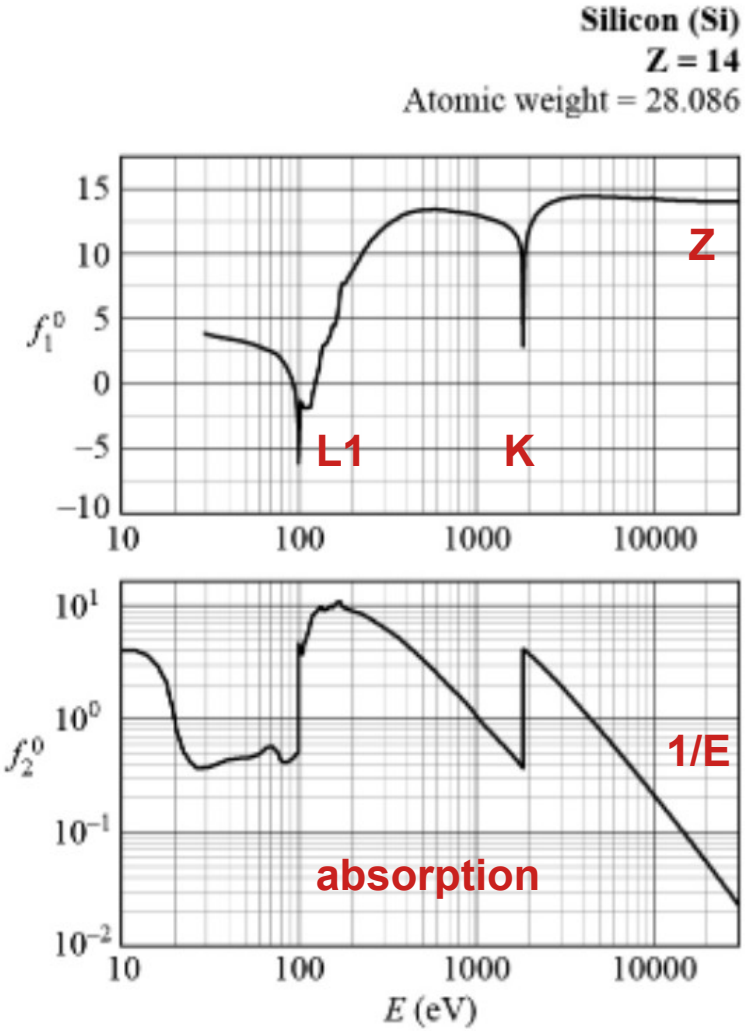
$r_e = e^2/mc^2 \sim 2.8$ fm

- $|f^0(\mathrm{w})|^2 \longrightarrow Z^2$ for $q \rightarrow 0$ and $\mathrm{w} \rightarrow \infty$

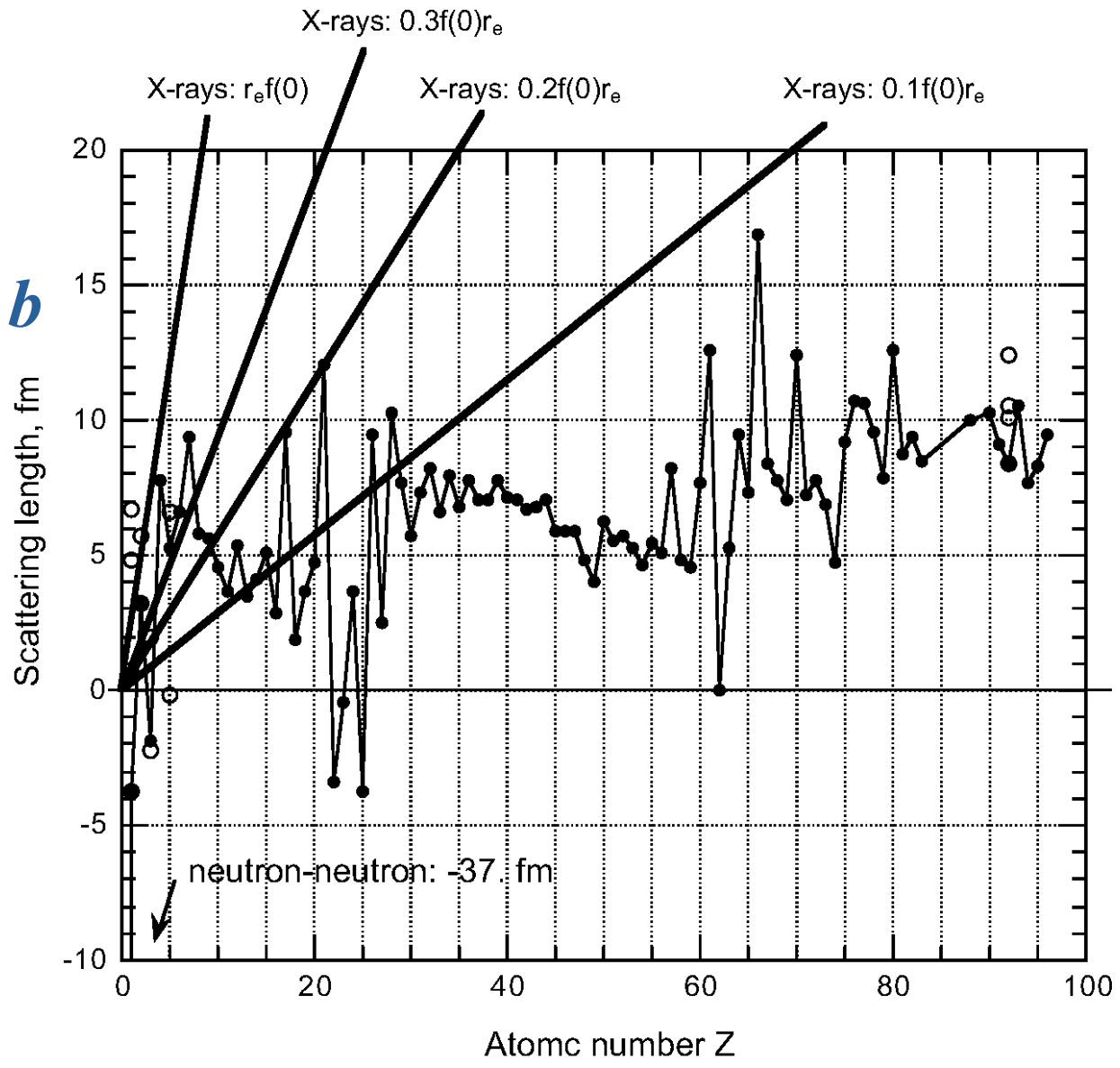
$$f^0(\omega) = \sum_{s=1}^{Z} \frac{\omega^2}{\omega^2 - \omega_s^2 + i\gamma\omega} = f_1^0(\omega) - i f_2^0(\omega)$$

- Some write

$$f(\omega) = Z - f_1(\omega) - i f_2(\omega)$$



Scattering length for atoms



X: Increases *vs* number of e⁻

n:

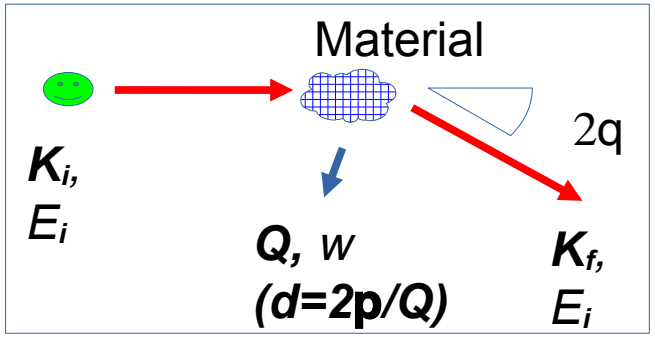


Particle-matter interaction (coh)

Selections rules

$K_f = K_i + Q$

$E_f = E_i + w$



Bragg's law (diffraction on structure – atoms separated by distance *d*)

$$n\lambda = 2\pi / K_i = 2d\sin\theta$$

Scattering law (intensity per solid angle and energy, dynamics) DDSC

$$\frac{d^2\sigma}{d\Omega\, dE_f} = \frac{K_f}{K_i}\left[\frac{\sigma}{4\pi} S(Q,\omega)\right]$$

$$\frac{k_2}{k_1} r_e^{\,2} \left|\varepsilon_2^{\,*} \bullet \varepsilon_1\right|^2 S(\mathbf{Q},\omega) \quad \text{per } e^-$$

n: Squires, *Th. Neut. Scatt.*

X: Schulke, *Inel X Scatt.* (Thomson scatt.)

Dynamical structure factor *S(Q,w)* is characteristic of each material
Reflects ordering of matter (atom/molecule positions – movements - domains)

Total scattering probability

The total scattering cross section is given in (W,E_f) space, but *S* is given in (q,w)
A variable change must be done for the integration (Jacobian).

We like to play games in (q,w) space

$$\frac{d\Omega}{d\theta} = -2\pi sin\theta$$
$$\frac{dq}{d\theta} = -\frac{k_i k_f sin\theta}{q}$$

Effective cross section in *(q,w)* space

$$\hat{\sigma} = \sigma \iint \frac{S(q,\omega)q}{2k_i^2} dq d\omega$$

V.F. Sears. *Adv. Phys.*, 24, 1, 1975.

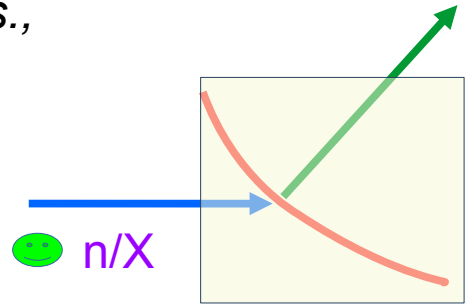
Probability to transmit

$$p = e^{-\rho \hat{\sigma} x}$$

Scattering distribution

S(q,w)

"probability" to scatter, preferably where *S* is large

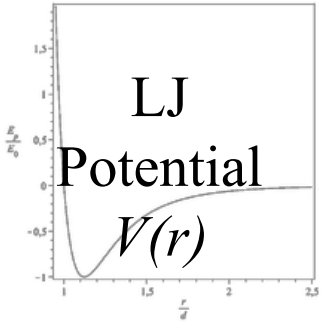


How do we estimate *S(q,ω)* ?

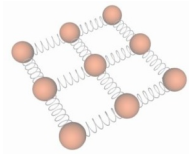
Harmonic potential

- The potential is related to the force acting on atoms as:

$$\boldsymbol{F} = -\nabla V$$



- Considering spring chains, we also have:



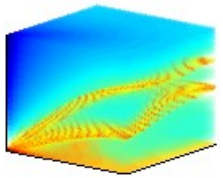
$$\boldsymbol{F} = -\boldsymbol{\Phi}.\boldsymbol{u} \quad i.e. \quad \boldsymbol{\Phi} = -\nabla F \text{ (Hooke's law, } \boldsymbol{u}=\boldsymbol{r}-\boldsymbol{r_0})$$

- So, in short, $\boldsymbol{\Phi} = \nabla^2 V$ is the *curvature* of the potential.
- If the potential is harmonic $V \sim \frac{1}{2}\boldsymbol{\Phi} u^2$ and $\boldsymbol{\Phi}$ is constant.
- Setting $u(r,t)=u_0\, e^{i(k.r-\omega t)}$ we get the tensorial form:

$$D_{\alpha\beta} = \nabla_{\alpha\beta}^2 V / \sqrt{M_\alpha M_\beta} = \Phi_{\alpha\beta}\, e^{ik.r} / \sqrt{M_\alpha M_\beta}$$

- And $\ddot{u}_\alpha = -\Sigma_\beta D_{\alpha\beta} u_\alpha$. Eigenvalues of D are $\omega(k)$

Phonons: small displacements



- Let us consider a perfect crystal, at very low temperature, in the harmonic approximation close to its equilibrium. The dynamical matrix *D* is defined as [***V***: potential, ***M***: mass, ***u***: periodic motion]:

$$D_{\alpha\beta} = \frac{1}{\sqrt{M_\alpha M_\beta}} \frac{\partial^2 V}{\partial u_\alpha \partial u_\beta} \qquad u_\alpha = U_\alpha\left(\vec{k}\right) e^{i\left(\vec{k}.\vec{r} - \omega t\right)}$$

- ***Step1***: Compute the potential and forces $F = -\nabla V \rightarrow \partial^2 V \rightarrow D_{\alpha\beta}$ with *ab-initio*/DFT code when displacing iteratively all atoms in the structure by d*u*.

- ***Step2***: Diagonalize the dynamical matrix at any HKL. Eigen-modes provide the phonon energies/polarisations at ***k***.

K. Parlinski, Z. Q. Li, and Y. Kawazoe, *Phys. Rev. Lett.* **78**, 4063 (1997)
D. Alfé, *Comp. Phys. Comm.* **180**, 2622-2633 (2009)

Vibrational intensity



$$n(\omega_j) = \frac{1}{e^{\omega_j(Q)/k_B T} - 1}$$

Population of the vibrational states vs *T*.

$$W_j(Q) = \frac{|\vec{Q}.\vec{e}_j|^2}{2\omega_j(Q)}\left(2n(\omega_j)+1\right)$$

Debye-Weller function ~ $\langle Q^2u^2 \rangle$

$$F_j(\vec{Q}) = \sum \frac{b_{coh}^{ref}}{\sqrt{m}} e^{-W_j(Q) - i\vec{Q}.\vec{r}} \vec{Q}.\vec{e}_j$$

Form factor of the vibration

Coherent inelastic "intensity" (Ta-Dam !)

$$S(Q,\omega) = \sum \left(n(\omega)+1\right)|F_j(Q)|^2 \frac{4\omega\Gamma/\pi}{\left(\omega^2 - \omega_j(Q)^2\right)^2 + 4\omega^2\Gamma^2}$$

SOLEIL II
La science éclaire l'avenir
Science lights up the future

What about powders ?



- $S(q,w) \sim S_{hkl} \mid F(hkl) \mid^2 d(w)$ pure elastic
- The intensity on a Debye-Scherrer cone is

$$\sigma_{\text{cone}} = \frac{V}{V_0^2} \frac{\lambda^3}{4 \sin\theta} \sum_Q |F(Q)|^2.$$

Can compute the structure factors F^2 with **cif2hkl**

<https://gitlab.com/soleil-data-treatment/soleil-software-projects/cif2hkl>

Works for *n* and *X* *(based on FullProf/CrysFML)*.

Thermodynamics from vDOS

- Density of States (vibrational)

$$g(\omega) = \sum \delta(\omega - \omega(k)) = \text{proj}(\omega(k))$$

- Thermodynamic quantities



Mean squared displacement

$$\langle u^2 \rangle = \frac{3\hbar}{3M} \int_0^\infty \frac{2n(\omega)+1}{\omega} g(\omega) d\omega$$

$$U = U_0 + \int \frac{\omega g(\omega)}{2} d\omega + \int \frac{\omega g(\omega)}{e^{\omega/k_B T} - 1} d\omega$$

Internal energy

$$S = \int \frac{\omega}{T\left(e^{\omega/k_B T} - 1\right)} - k_B \ln\left(1 - e^{\omega/k_B T}\right) d\omega$$

Entropy

$F = U - T S$

Helmoltz free energy

$C_v = dU/dT$

Heat capacity

Approximate contributions

- gDOS $$g(\omega)=\frac{\omega\left[1-e^{-\hbar\omega/k_BT}\right]}{q^2}S_{inc}(q,\omega)\simeq\lim_{\omega\ll k_BT}\frac{\omega^2}{q^2}S_{inc}(q,\omega)$$

Price *J. Non Cryst Sol* **92** (1987) 153
MCBF *J. Mol. Struct.* **250** (1991) 213

- Use the "incoherent approximation"

$$S(\vec{Q},\omega)=\frac{\sigma_{\text{inc}}}{4\pi}\frac{N}{2\pi\hbar}\int_{-\infty}^{\infty}\mathrm{d}t\mathrm{e}^{-\mathrm{i}\omega t}\mathrm{e}^{-2W(\vec{Q})}\mathrm{e}^{\langle(\vec{Q}\cdot\vec{u}_j)(\vec{Q}\cdot\vec{u}_j(t))\rangle}$$



- Expanded in Taylor series, where the 0-order contains the vDOS $g(w)$, and further terms are auto-convolutions. [Schober *JNR* (2014), Sjölander *Arkiv Fysik* (1958)]



- Provides $S_{inc}(q,w)$ from $g(w)$.
- Estimate the coherent contribution (*caution* !)

$$S_{coh}(q,w)\sim S_{inc}(\,q/\sqrt{S(q)}\,,\,w)\,.S(q)$$ [Sköld *PRL* (1967)]

What about X-rays ?



- Equations are based on σ for single e^- ($r_e^2 = 0.67$ b)
- No incoherent scattering, but Compton/Fluorescence.
- Must multiply r_e^2 in the DDCS by $|f^0(\omega)|^2 \rightarrow Z^2$

$$\sigma_x \rightarrow 0.67\ Z^2$$

Get *f* for all atoms at:

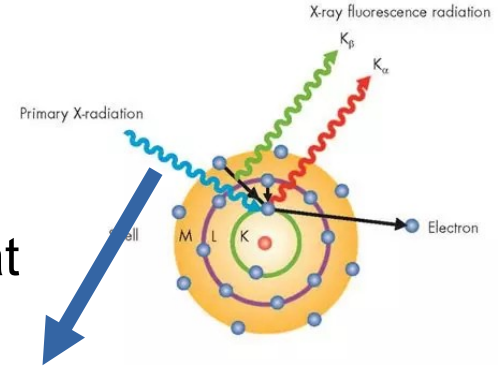
https://physics.nist.gov/PhysRefData/FFast/html/form.html

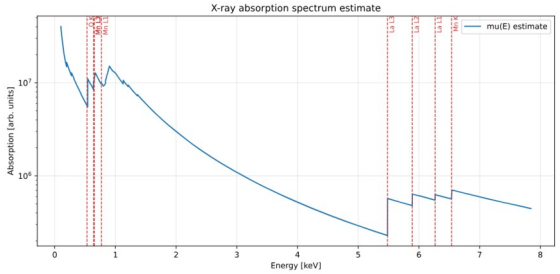
https://github.com/tschoonj/xraylib

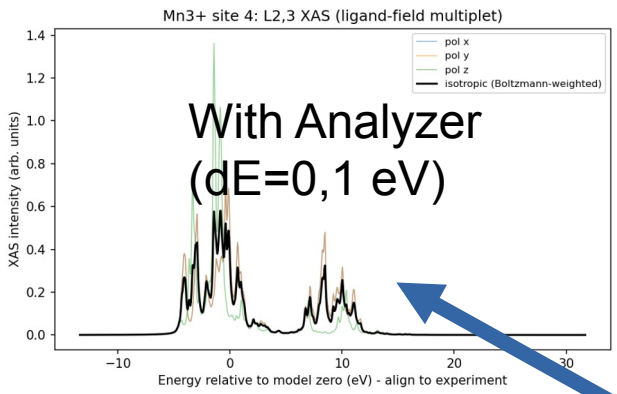
Only for X: Photo-emission

Most X-ray beam-lines are equipped with fluorescence detectors.

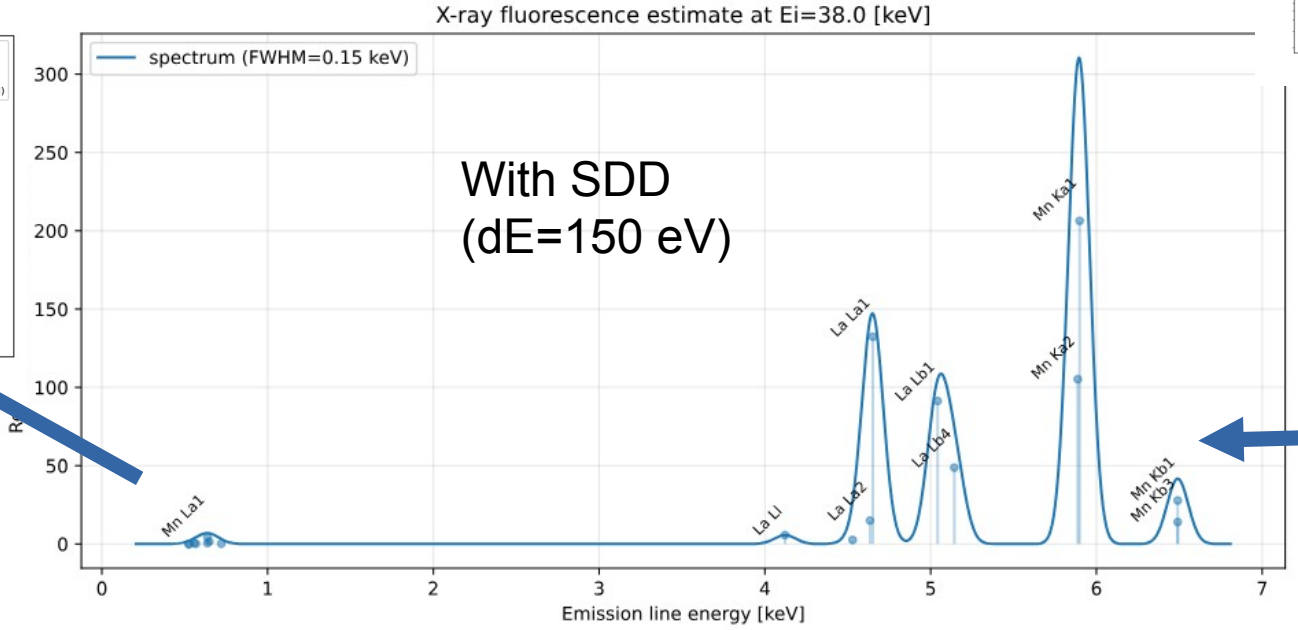
- Photons eject electrons in an absorption process.
- The holes are filled by an electron that “falls down” while emitting a 2^{dary} photon at lower energy.
- Generates a 1D energy spectra (e.g. 1-40 keV) [XrayLib]
- 3d/4d/f transition metals exhibit a fluorescence fine structure.



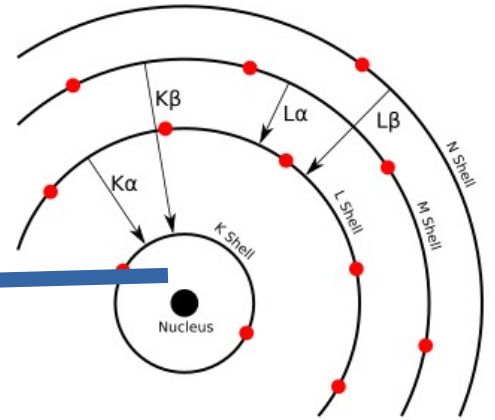




With Analyzer
(dE=0,1 eV)



With SDD
(dE=150 eV)



PS: epithermal neutrons can eject photon-neutrons.

SOLEIL II
La science éclaire l'avenir
Science lights up the future

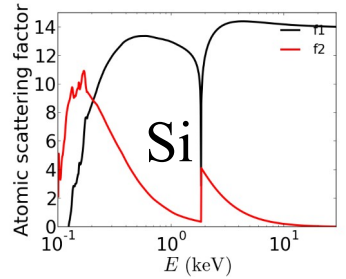
Computing absorption



- Both *n* and *X* can be absorbed. $I/I_o \sim e^{-\rho\sigma x}$
- **Neutrons**: few resonances in the cold-thermal-hot range. $\mu = \rho\sigma$
- **X-rays**: many resonances/abs. edges in 0-100 keV. Allow to discriminate elements. Computed from the imaginary part of the atomic scattering factor f_2 (or f'').

$$\mu = \frac{\rho N_a}{M_a} 2 r_e \lambda f_2$$

Look at <http://gisaxs.com/index.php/Absorption_length> for X-rays



Coherent / incoherent scattering

- Coherent scattering channels both active with *n/X*
- Model waves in materials (phonons, magnons, spinons, …)
- Incoherent scattering related to e.g. diffusion

  Not active with X-rays (but has Compton/Fluo scatt.).



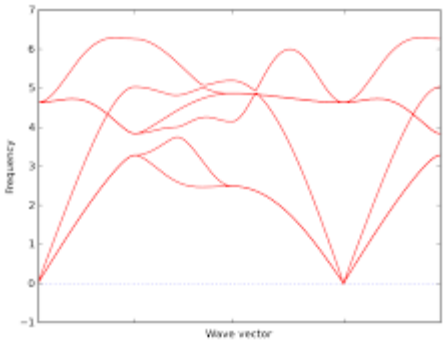
Applications

Where all this is used

SOLEIL II
La science éclaire l'avenir
Science lights up the future

Existing software

- **PhonoPy**+VASP/QE/Abinit...: free, powerful.
- **ASE**: free, limited for phonons. Vast choice of DFT code. EU project, links with PhonoPy.
- MedeA+VASP/MOPAC: commercial
- Material Studio+CASTEP: commercial





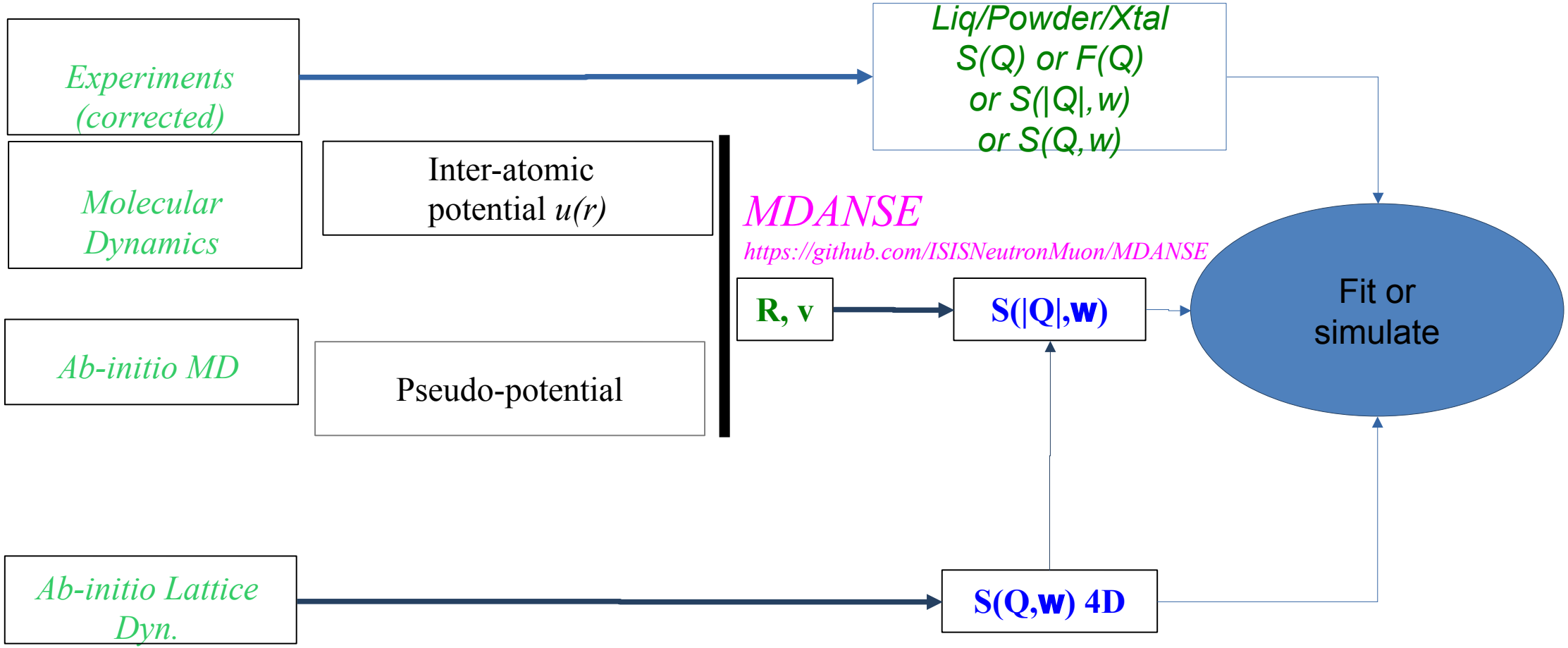
Existing databases

Before starting any new study from scratch, you should consider to inquire for existing structures and vibrational information from **databases** at:

- https://alexandria.icams.rub.de/
- https://optimadeclient.materialscloud.io/ (catalog of catalogs)
- https://www.materialscloud.org/home (QE team)
- https://nomad-lab.eu/nomad-lab/ (simulated structures)
- https://next-gen.materialsproject.org/ (simulated structures)
- https://www.crystallography.net/cod/ (sim and exp)
- https://github.com/atztogo/phonondb (phonopy db)



How to get *S(q,w)* data sets



S(q,w) from experiments

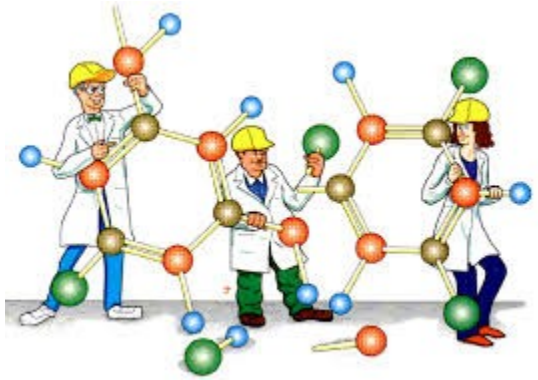
- Reduce the experimental data, correct it for e.g. empty cell, parallax, detector efficiency, …
- Correct for absorption, multiple-scattering, …
- Estimate incoherent scattering and subtract to get the « pure » coherent part.



S(q,w) from MD

- Set the system box, (pseudo) potential/FF
- Equilibrate, and couple to thermostat (NVT)
- Run calculator in MD mode (NVE)
- Convert trajectory (**R**,t) into S(|Q|,w) [MDANSE]

  ➔ Analysis/Scattering/DCSF and DISF → .nc/.h5 file
- Explore file to search for [q,w,sqw] (ncview,hdfview/silx)

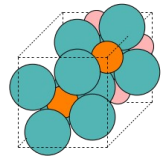


S(q,w) from Lattice Dynamics

- Create a crystal, calculate displacements or so.
- Assemble the ‘phonon’ representation.
- Diagonalize the dynamical matrix.
- Compute the vDOS
- Estimate incoherent scattering
- Evaluate *S(Q,w)* on powder average or *(h,k,l,w)*

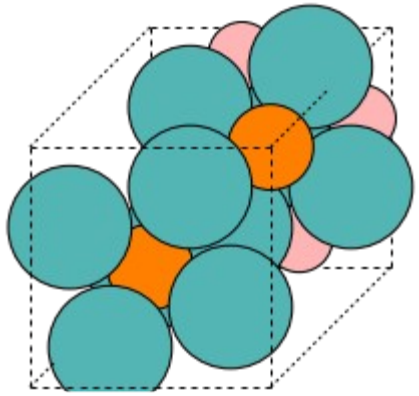


An example: $MoPB_2$



A tetragonal I4/mcm (140) boride, which exhibits super-conducting properties, and is studied as well as a promising ceramic material.

Let's summarize the methodology to compute *S(**q**,w)*.



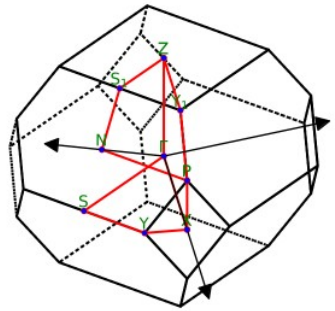
B4Mo10P2 BCT(a=6.00172, c=11.1709)

Import structure

- import **ase**
- atoms = ase.io.read(CIF) *# Alexandria 3477599*
- calculator = Espresso(...)
- atoms.set_calculator(calculator)

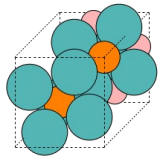
Optimize the structure

- opt = ase.optimize.BFGS(atoms)
- opt.run()



Bravais lattice

$MoPB_2$ electronic structure

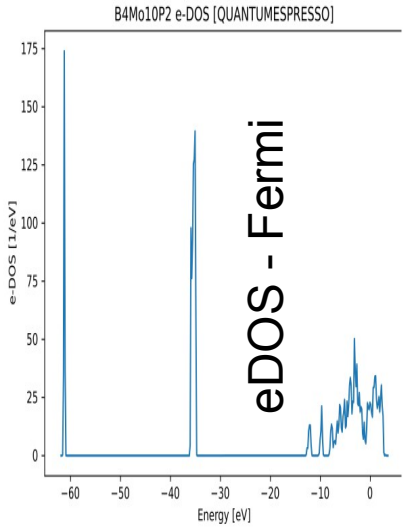


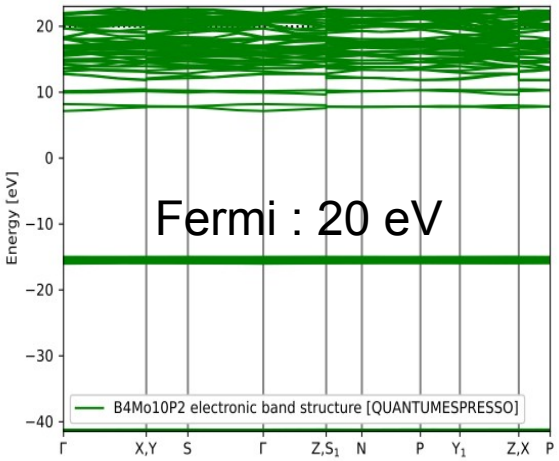
Compute the e- DOS

- atoms.calc.get_dos()

or

- dos=ase.dft.dos.DOS()



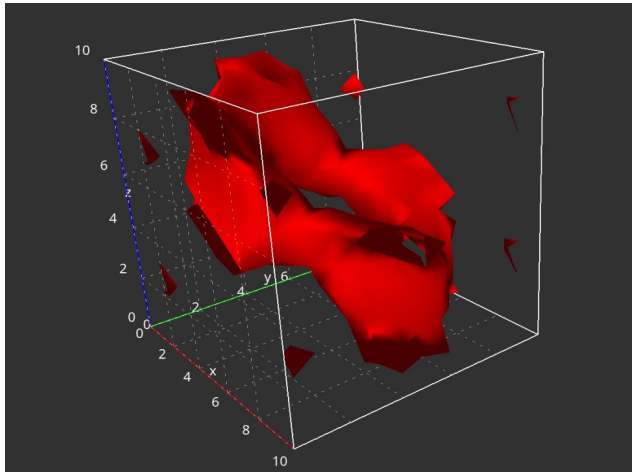


Compute e- band structure

- atoms.calc.set(kpts=atoms.cell.bandpath())
- bs=atoms.calc.band_structure()

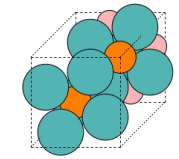
Compute e- band structure in 3D

- grid = ase.dft.kpoints.monkhorst_pack([10, 10, 10])
- bp = ase.dft.kpoints.BandPath(atoms.cell, kpts=grid)
- atoms.calc.set(kpts=bp)
- bs = atoms.calc.band_structure()



SOLEIL II
La science éclaire l'avenir
Science lights up the future

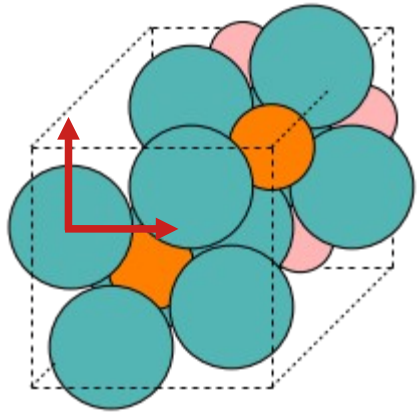
$MoPB_2$ vibrational dispersions



We use Phonopy (with ASE).

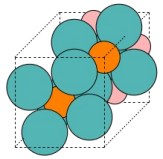
The iterative calculation of the material energy and forces *vs* atom displacements is made easy.

Compute the phonons



- import **phonopy**
- ph = phonopy.**Phonopy**(atoms, supercell)
- ph.generate_displacements()
- supercells = list(ph.get_supercells_with_displacements())
- for scell in supercells:
- forces = scell.get_forces()

$MoPB_2$ vibrational dispersions



Compute the vDOS

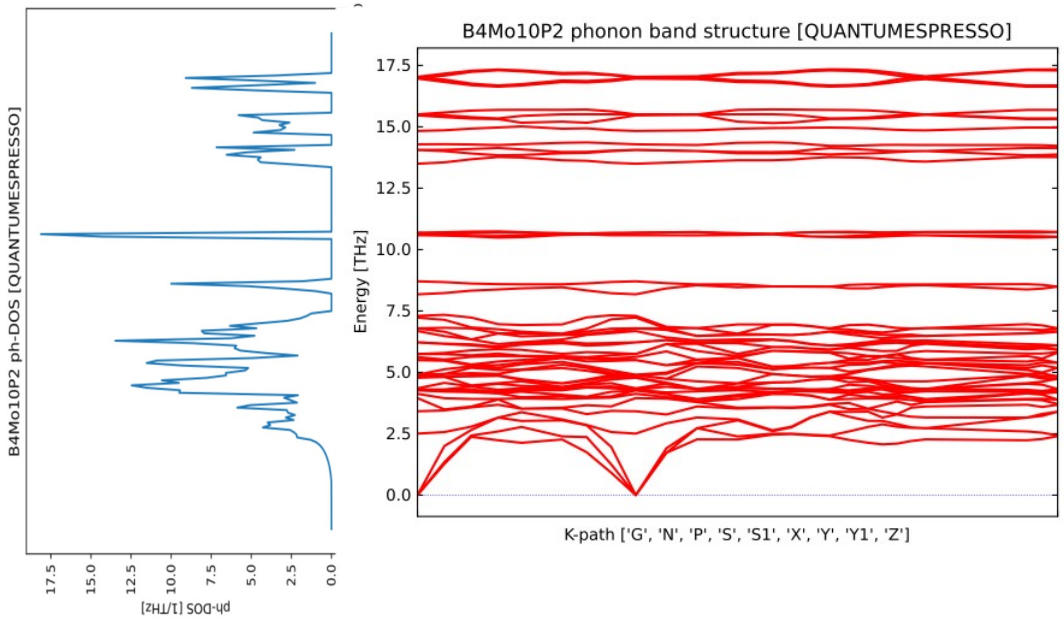
- ph.auto_total_dos()
- dos = phonon.get_total_dos_dict()

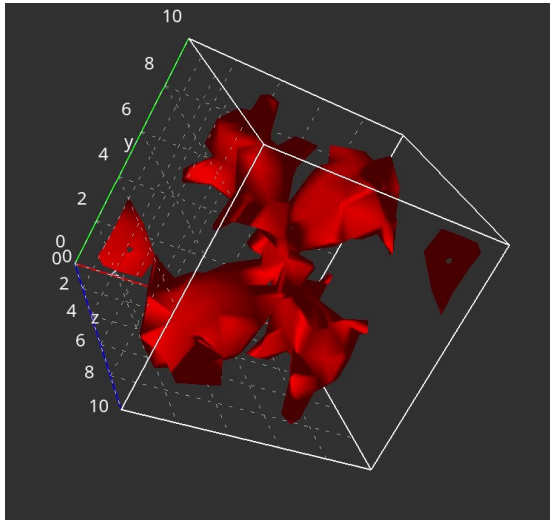
Compute the phononic band structure

- bandpath = atoms.cell.bandpath()
- ph.run_band_structure([bandpath.kpts])

Compute the phononic band structure in 3D

- bandpath = ase.dft.kpoints.monkhorst_pack(dims)
- ph.run_band_structure([bandpath])





Automating most of it ?

Material-modelling: An automatic service/tool to compute electronic and vibrational properties :

- https://gitlab.com/soleil-data-treatment/soleil-software-projects/material-modelling-service
- https://data-analysis.synchrotron-soleil.fr/computing/material-modelling.html

Calculation is « as fast as possible » to provide quick estimates for e.g. neutron/X-ray experiments.
Relies on ASE, QE and PhonoPy.

Provides :

- Electronic ground state
- Electronic density of states, dispersion (band-path), dispersion in HKL
- Vibrational density of states, dispersion (band-path), dispersion in HKL
- X-ray Fluorescence and absorption edges
- Diffractogram

Outputs data in many formats, including HDF5 and McStas/McXtrace S(q,w) 2D/4D.

Material Modelling

Used at SOLEIL for fast material estimates (ARPES, Fluo/XAS, eDOS)

Comes with FEFF and Quanty helpers for EXAFS and XPS.

Still working on spin-polarization and Hubbard correction.

Basis: human
Appearance: AI



Material Modelling Service

This service computes electronic and phononic properties of materials using first-principle theory (DFT - *ab-initio*). The only requirements are a name and a crystal structure file (CIF/PDB/POSCAR, etc.) supported by ASE.

You can choose the specific calculator to use and the computational steps to perform.

Your User ID * farhi

Calculator: QuantumEspresso: Plane-wave pseudopotential code/ASE. Pseudo: SSSP Efficiency

Smearing: Auto/semiconductor (smearing 0.1 meV)

Computations:

- [x] Electrons (DOS + band structure)
- [x] Electrons 3D (full BZ dispersions)
- [x] Phonons (DOS + band structure)
- [x] Phonons 3D (full BZ dispersions)
- [x] Diffraction I(q) (neutrons + X-rays)
- [x] X-ray fluorescence estimate
- [x] X-ray absorption estimate (XANES)
- [ ] Cell optimisation (generate new CIF)

Select any combination. Electrons is checked by default. Phonons/diffraction require a converged ground state; 3D steps are longer.

Material structure * Browse... LaB6.cif

Max 100kB

Compute!

Material Modelling: examples

Examples for full computations n/X with full data format exports

Service at https://data-analysis.synchrotron-soleil.fr/material-modelling/

Al 2 min
https://data-analysis.synchrotron-soleil.fr/material-modelling/data/Al.cif-QuantumEspresso-farhi_test-20261010-155744/

LaB_6 15 min
https://data-analysis.synchrotron-soleil.fr/material-modelling/data/LaB6.cif-QuantumEspresso-farhie-20261101-190845/

PB_2Mo_5 50 min
https://data-analysis.synchrotron-soleil.fr/material-modelling/data/B4Mo10P2.cif-QuantumEspresso-farhi-20261101-205606/

What to do with this ?

Create annotated data for AI training

McXtrace

https://mcxtrace.org/

McStas

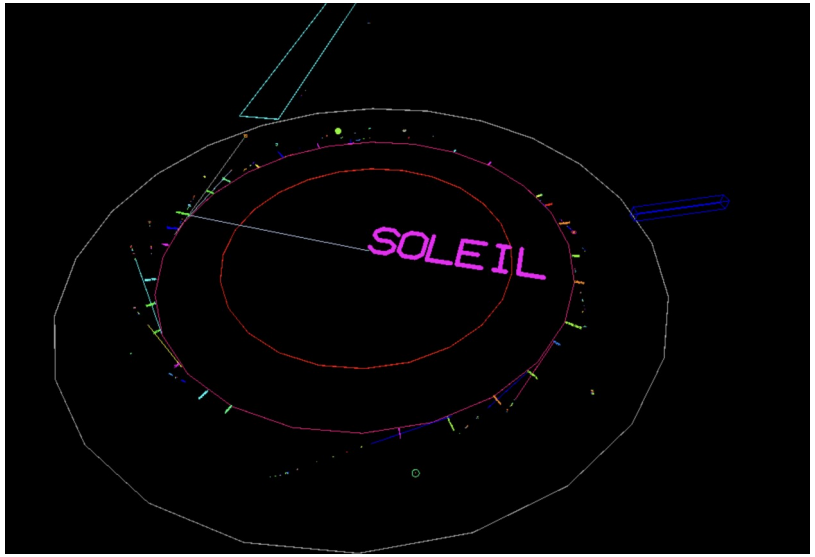
https://mcstas.org/

Realistic « AI » datasets:

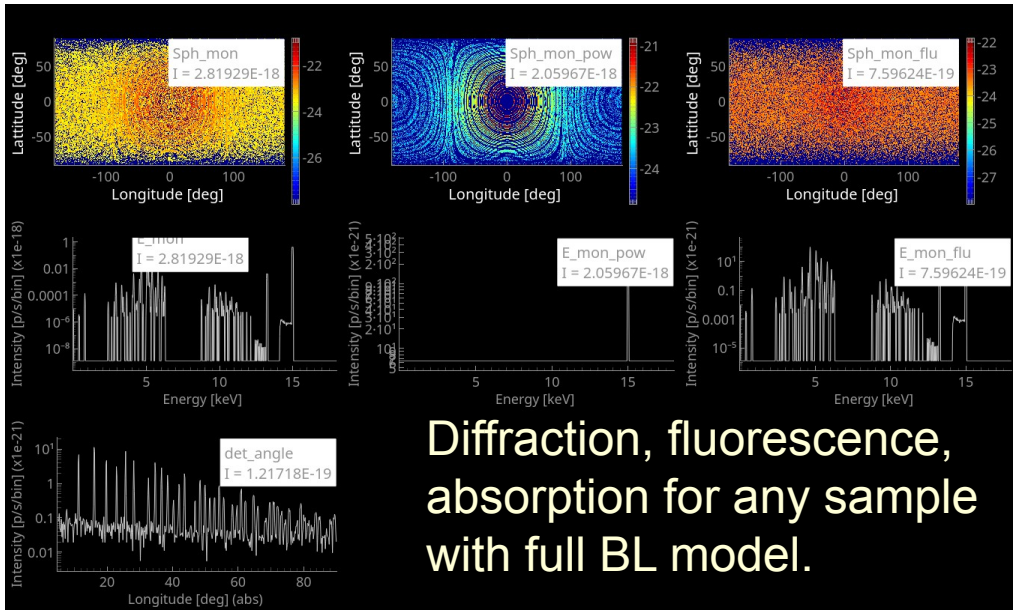
- Full BL/instrument models with samples, and « background noise »
- **McXtrace/McStas** (for X and *neutrons*).
- We currently have 18/29 SOLEIL BL's: photon sources, optics, samples, detectors.

Applications:

- Surrogates, inverse problems, parameter optimisation, data filtering.
- BL control prototype from LLM (with samples).
- BL automatic alignment (on sample or detector).
- LLM based agent for data processing.



Digital twins → AI training data



Diffraction, fluorescence, absorption for any sample with full BL model.

McStas / McXtrace sample models

Aside all particle **sources**, **optics** and **detectors**, both McXtrace and McStas provide:

- Diffraction (powders and single-crystals)
- Small angle scattering (most SASView models)
- Inelastic scattering (Sqw)
- Fluorescence/absorption, Compton, Rayleigh (for X)
- Incoherent scattering (for neutrons)
- And many more...

Numerous beam-lines/instrument models exist as examples, esp. **ILL** and **SOLEIL**.
This allows to model most “virtual experiments” around, to create extensive realistic datasets. Contribute !

Thanks for listening

Emmanuel.farhi@synchrotron-soleil.fr

https://data-analysis.synchrotron-soleil.fr